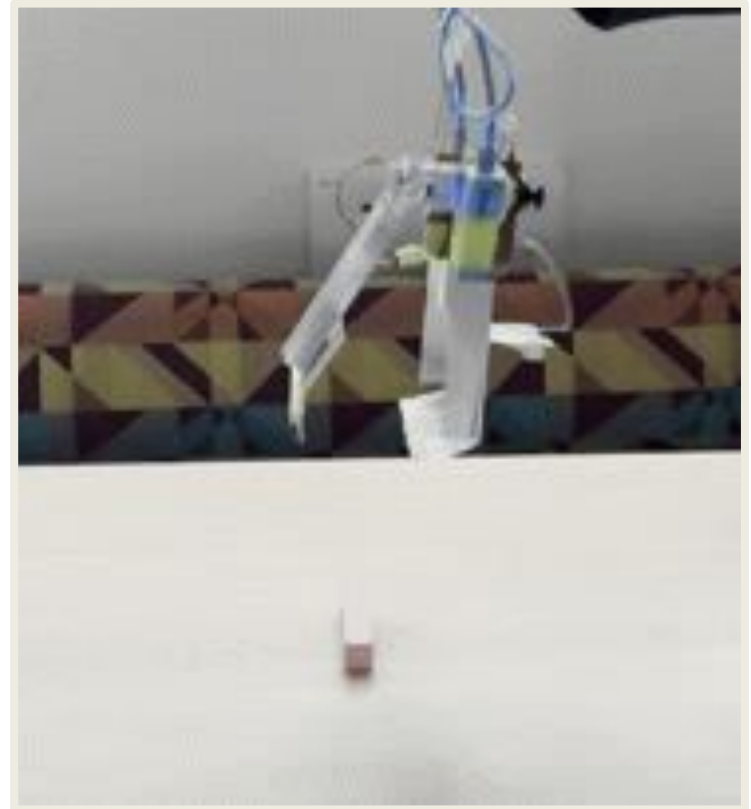


THE SWEEPER

Autonomous Retrieval
System

Team B5



CONTEXT

- A unidentified flying object (UFO) crash has left **various items lost** in the Pacific Ocean.
- Due to unworkable conditions, **human recovery is not possible**.
- Objective: generate a **fully autonomous** recovery system.

TARGET DESIGN SPECIFICATIONS / COMPETITION STRATEGY

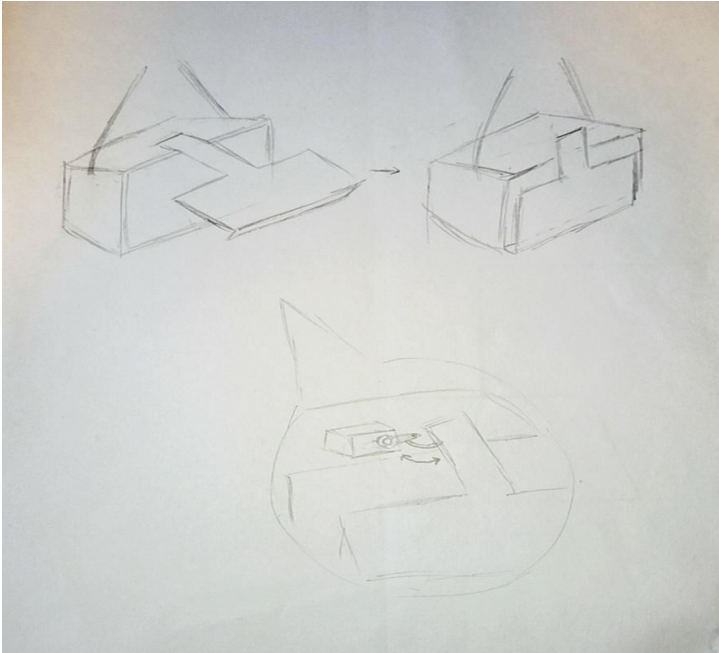
Design Aspect:

- Optimal claw shape and torque to pick up objects of varying shapes and sizes.
- Simple design for emergency replacements and adjustments.

During Competition:

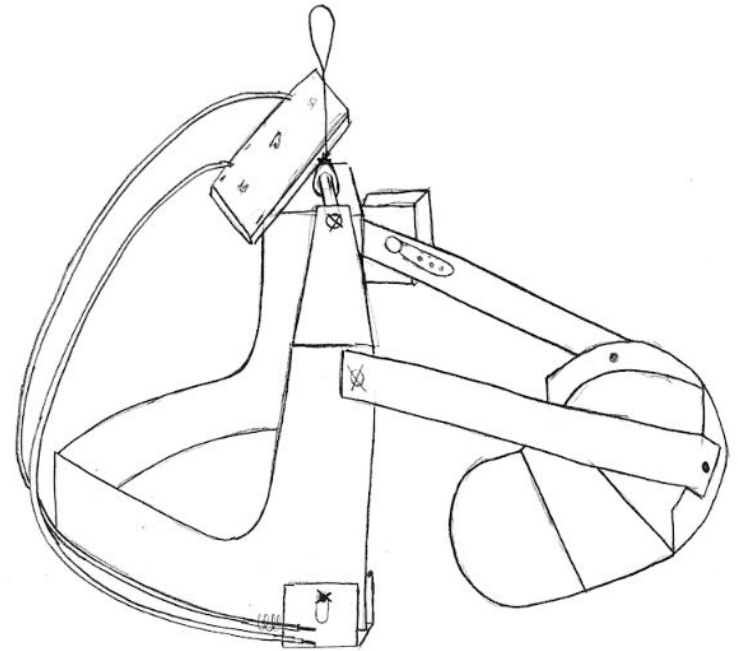
- Operator should move the crane slowly to avoid swinging.
- Callers should provide distinctive instructions for horizontal and vertical positions.

INITIAL SKETCHES



The Sweeper

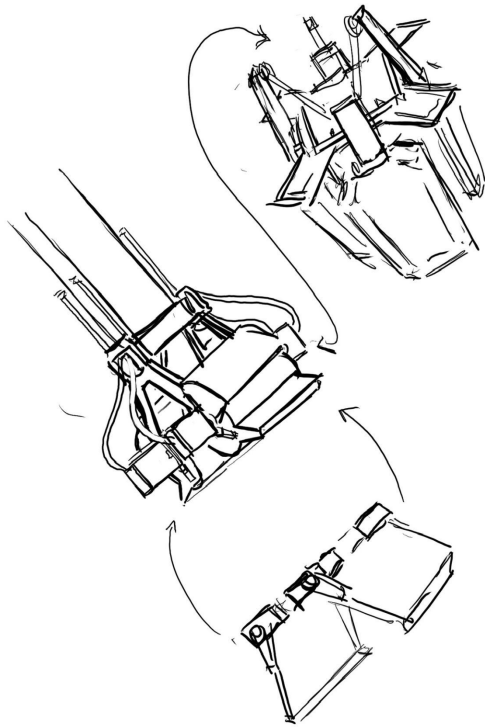
Uses a simple back and forth sweeping motion using the 180 degree servo motor range.



Shovel Rim Scooper

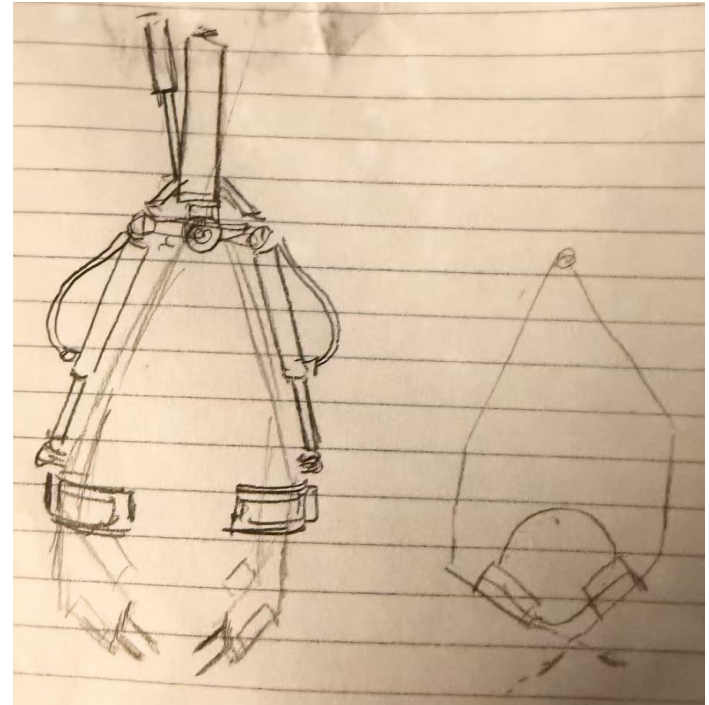
Upon contact with the ground, activates the scooper that pushes objects toward the rim.

INITIAL SKETCHES



Rotate Clip

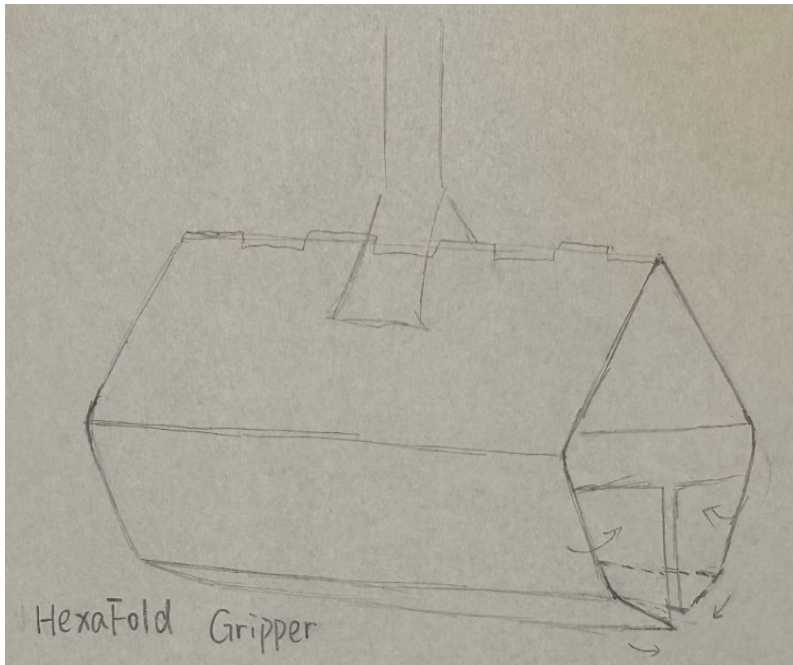
The object is clamped and moved by the rotation of two (or four) cylindrical structures.



Claw with Side Guards

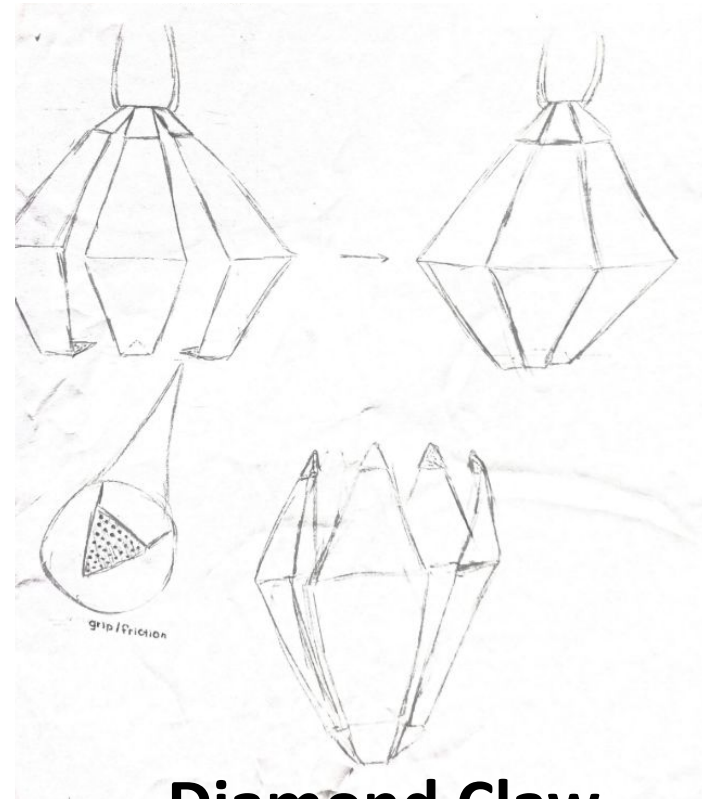
Featuring interlocking serrated edges, with small rectangular side guards on each side.

INITIAL SKETCHES



HexaFold Gripper

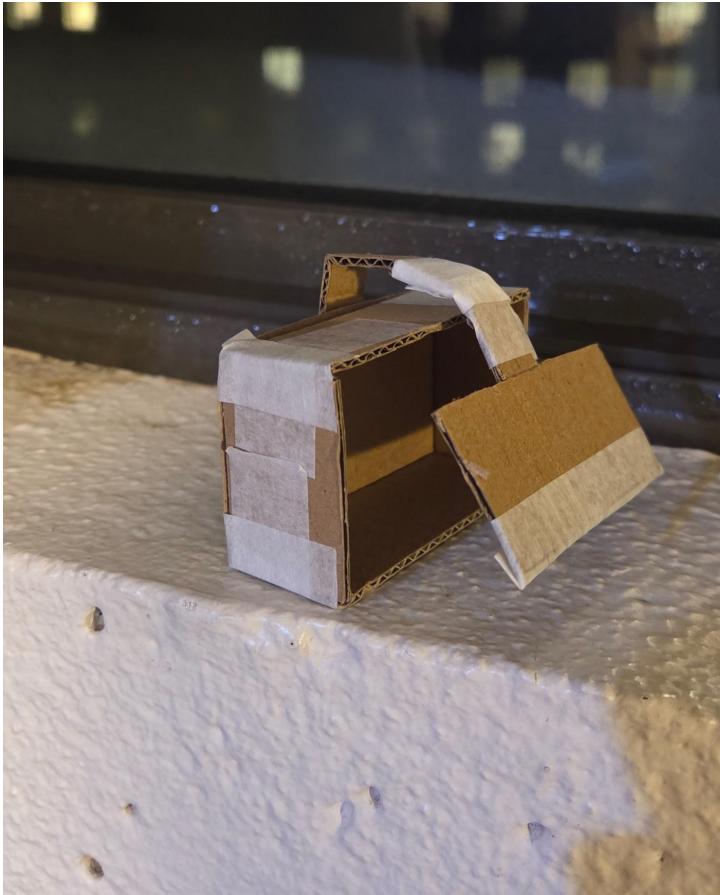
A multi-fold gripper that specializes in holding longer objects. Assisted by side stabilizers.



Diamond Claw

When closed, this four-pronged claw is able to fully encapsulate the object.

EARLY PHYSICAL PROTOTYPES

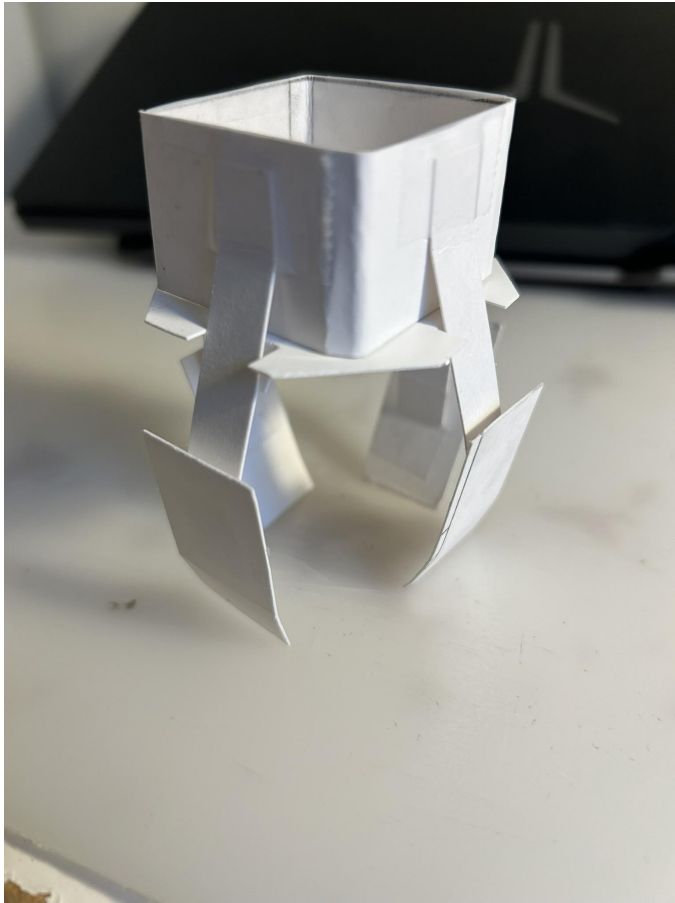


The Sweeper



Shovel Rim Scooper

EARLY PHYSICAL PROTOTYPES



**Rotate Clip Four Clip
Variant**

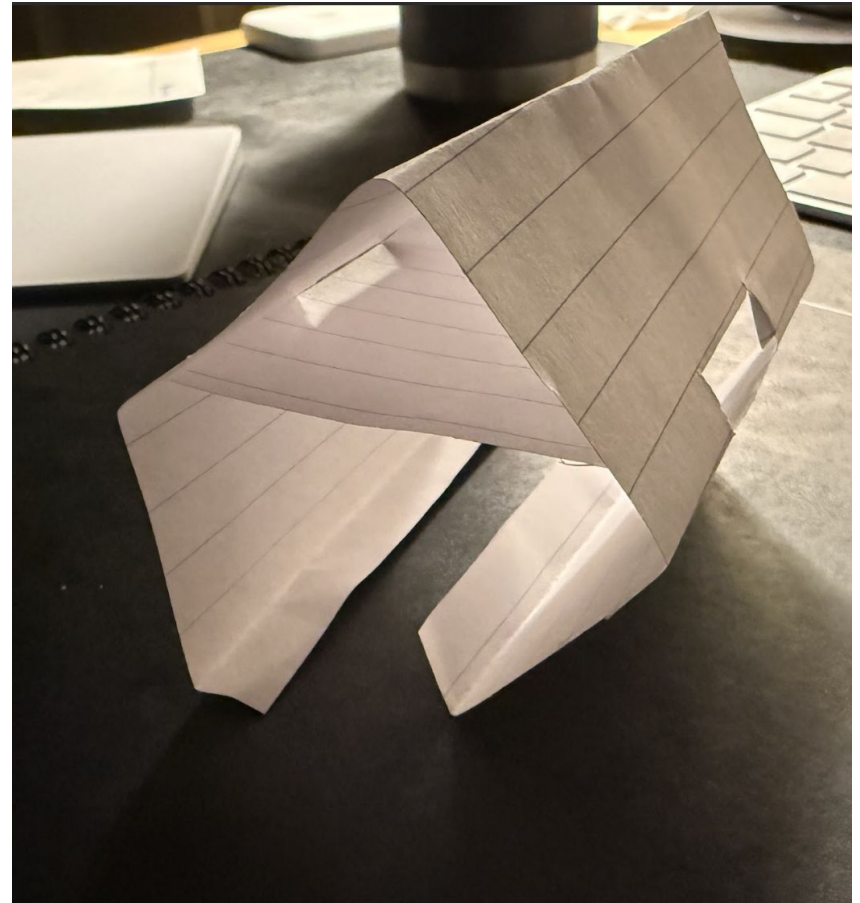


Claw with Side Guards

EARLY PHYSICAL PROTOTYPES



HexaFold Gripper



Dual Sided Claw

RISK ASSESSMENT

The following risks were identified:

1. Consistently sensing the ground and knowing when to close the claw
 - **Technical** & **Preventable**
2. Making our claw lighter with less material, but sacrificing structural integrity
 - **Technical** & **Strategic**
3. Ensuring the servo motor has enough torque to move the claw arm
 - **Technical** & **Preventable**
4. Reliably securing wires to the arduino kit components
 - **Technical** & **Preventable**
5. Communicating productively between crane operators and callers
 - **Operational** & **Preventable**

RISK REGISTER

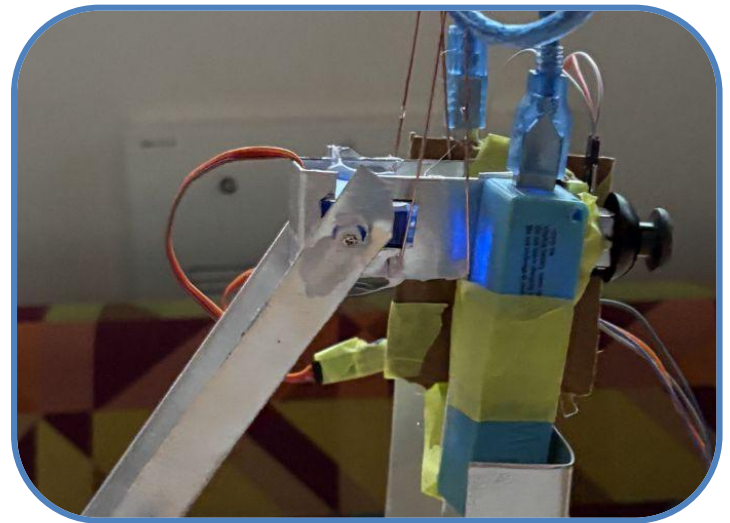
Risk	Severity (1-5)	Likelihood (1-5)	Rating	Action	Member Responsible (Initials)
1 - Sensing the Ground	5	2	10	Integrating code and physical claw & extensive testing.	ME
2 - Less Material	3	3	9	Test material load capacity.	WK
3 - Servo Motor Torque	5	3	15	Test various connection methods.	BZ, ML
4 - Wire Connections	2	3	6	Securing the wires with an adhesive (tape or glue).	AZ
5 - Crane Communication	3	3	9	Establish direction communication & practice.	EG, ZL

RISK MANAGEMENT & INVESTIGATIVE PROTOTYPES

To further mitigate our most highly rated risks, we conducted **two experiments** using **investigative prototypes**:

1 Reliably sensing the ground and knowing when to close the claw.

2 Servo motor maintaining enough torque to move the claw arm.



INVESTIGATIVE PROTOTYPES

1

Which Sensing and Trigger Mechanism provides the Most Reliable Detection of Ground Contact for Automatic Claw Action?

Approach: developed various code iterations and conducted small-scale tests to evaluate sensor accuracy and servo response at varying heights.

Compared Methods:

Detection:

2-wire contact sensing
versus sonar distance
sensing

Closing:

Timed automatic close
versus manual close
mechanism

Additional:

Use of joystick for
emergency release
override

Outcome: We selected sonar for a more reliable ground detection, and implemented a height threshold that triggers and holds automatic closing until the claw is lowered below a set release height. An emergency joystick override will immediately open the claw in the case of a missed retrieval.

INVESTIGATIVE PROTOTYPES

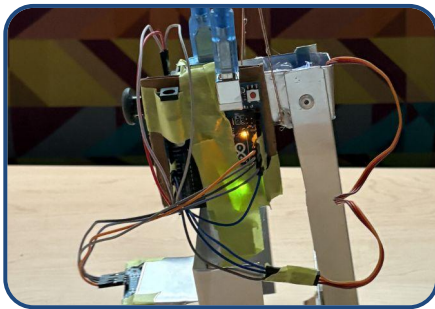
2

Which Claw Arm Geometry and Servo Placement Maximizes Torque for Effective Claw Closure?

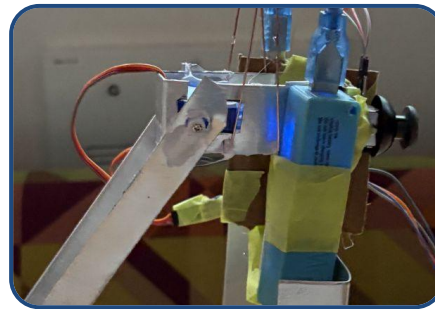
Approach: compared multiple arm geometries, materials, and servo mounting positions to evaluate greatest closing force

Compared Methods:

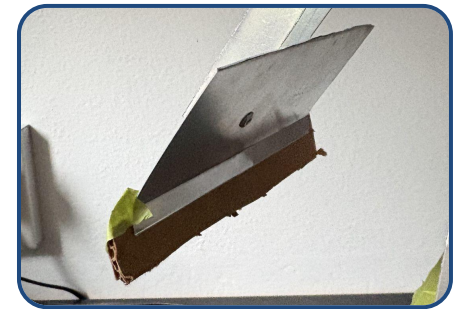
1. Top-mounted versus side-mounted arduino position



2. Varying angles of servo arm (to adjust closure of claw)



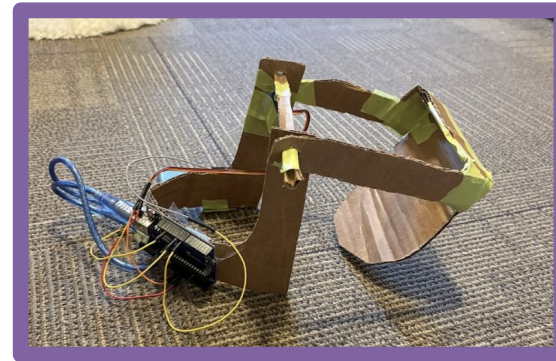
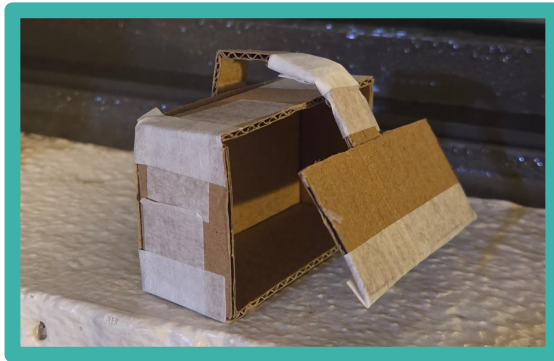
3. Cardboard versus aluminum usage



Outcome: We selected a top-mounted position for the arduino board and power bank, to improve the centre of gravity and overall balance, adjusted the servo arm angle to achieve effective open and closed positions, and opted to use aluminum over cardboard for a stronger hold.

DECISION MAKING PROCESS

- A **Weighted Decision Matrix** was used to compare our design options.
 - Evaluation criteria were based on key performance considerations.
- The **sweeping-style mechanism** outperformed traditional claws.
- Two concepts, **The Sweeper** and the **Shovel Rim Scooper** were selected for further prototyping.



- **The Sweeper** demonstrated superior overall performance and was selected for further development.

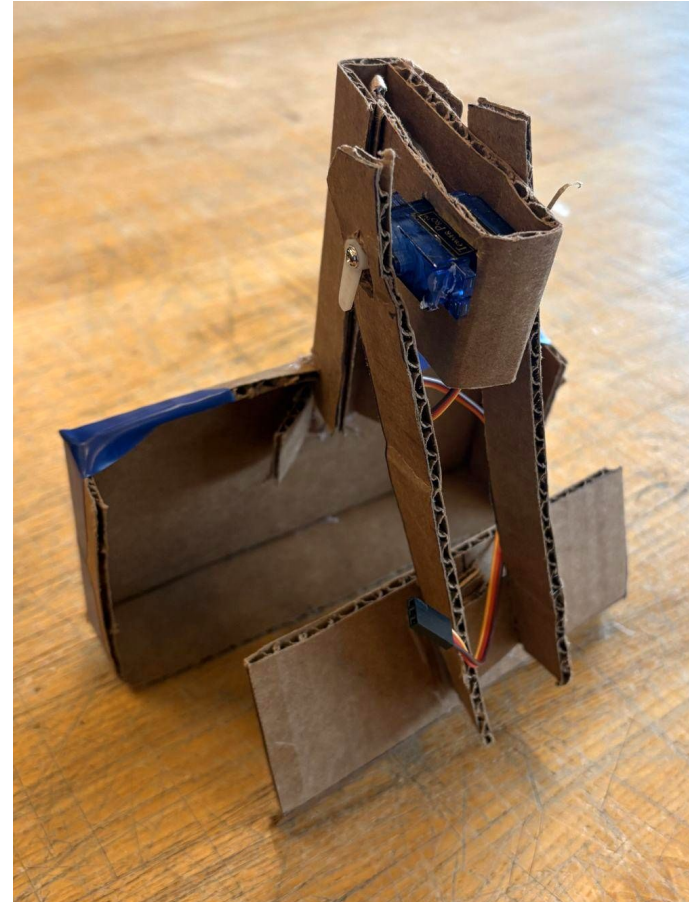
WEIGHTED DECISION MATRIX

Evaluation Criteria	Weight	Sweeper		Shovel Rim Scooper		Rotate clip		Claw with side guards		HexaFold Gripper	
		Score	Weighted	Score	Weighted	Score	Weighted	Score	Weighted	Score	Weighted
Maximum opening size	10%	9	0.9	9	0.9	6	0.6	5	0.5	8	0.8
Grasping ability	20%	8	1.6	6	1.2	7	1.4	8	1.6	6	1.2
Stability during lifting and transport	15%	7	1.05	7	1.05	5	0.75	8	1.2	6	0.9
Controllability of release	10%	8	0.8	8	0.8	8	0.8	6	0.6	8	0.8
Repeatability	15%	10	1.5	10	1.5	7	1.05	7	1.05	7	1.05
Robustness to object misalignment	10%	8	0.8	7	0.7	8	0.8	7	0.7	8	0.8
Complexity of sonar-servo control implementation	10%	9	0.9	9	0.9	7	0.7	6	0.6	7	0.7
Manufacturing complexity	10%	5	0.5	5	0.5	4	0.4	5	0.5	5	0.5
	100%										
Weighted Score			8.05		7.55		6.5		6.75		6.75

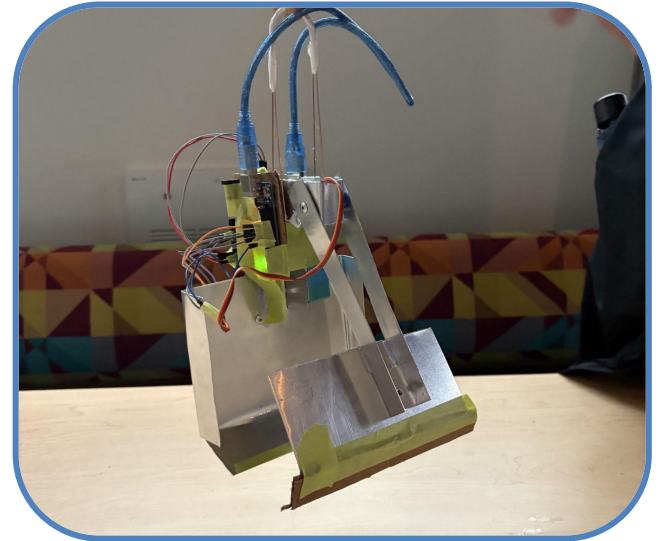
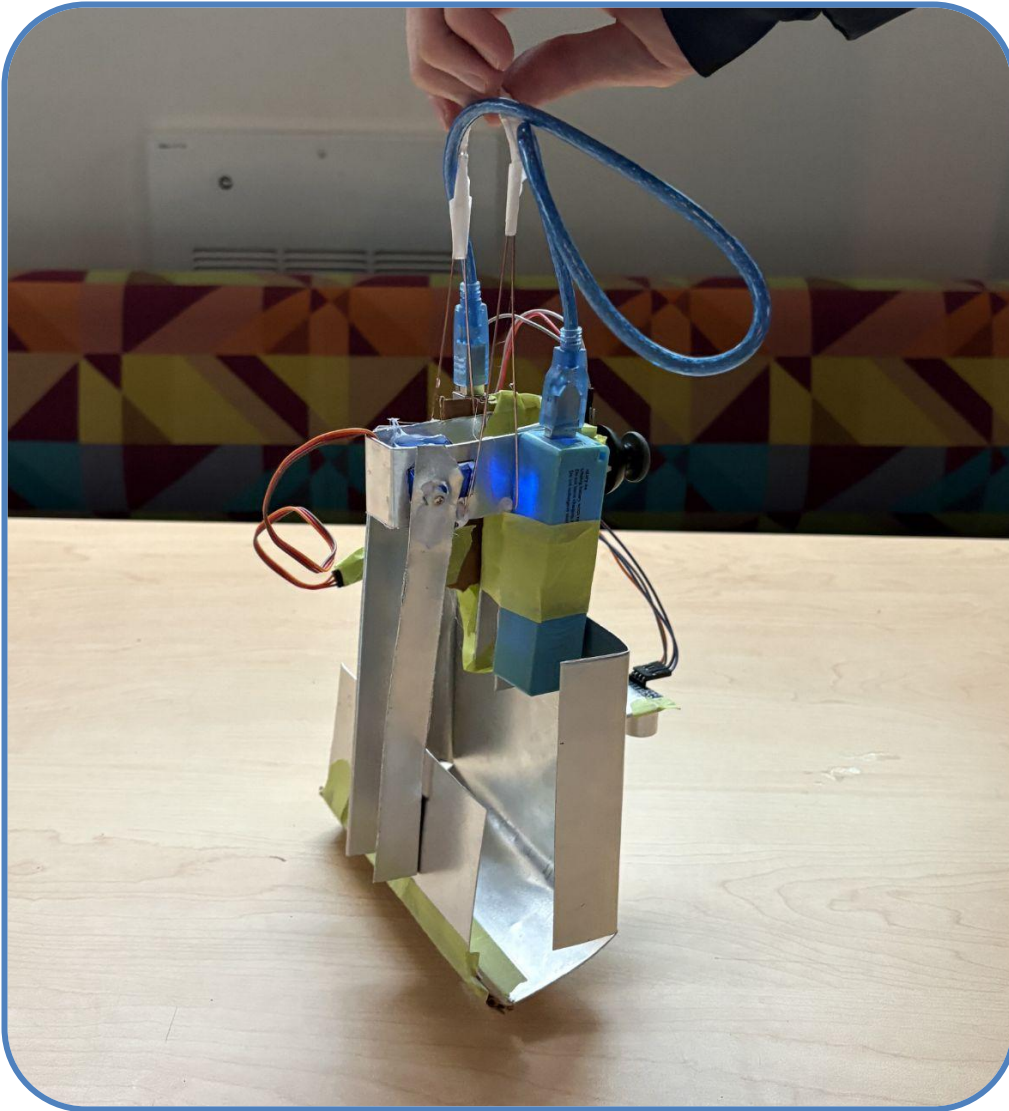
REFINEMENTS FOR FINAL DESIGN

Structural refinements were implemented to **The Sweeper** to improve functionality:

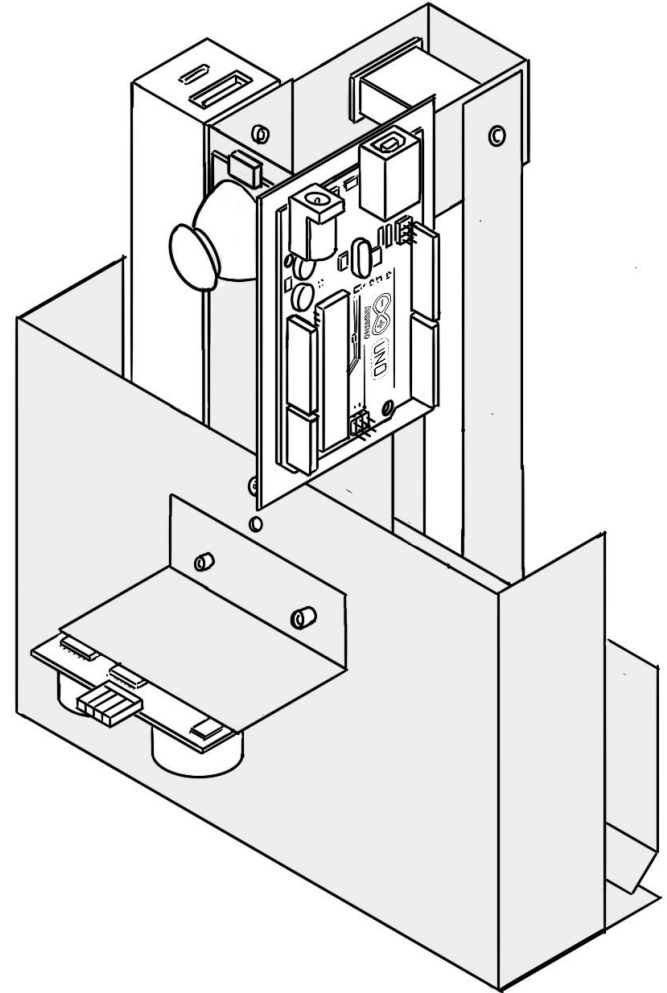
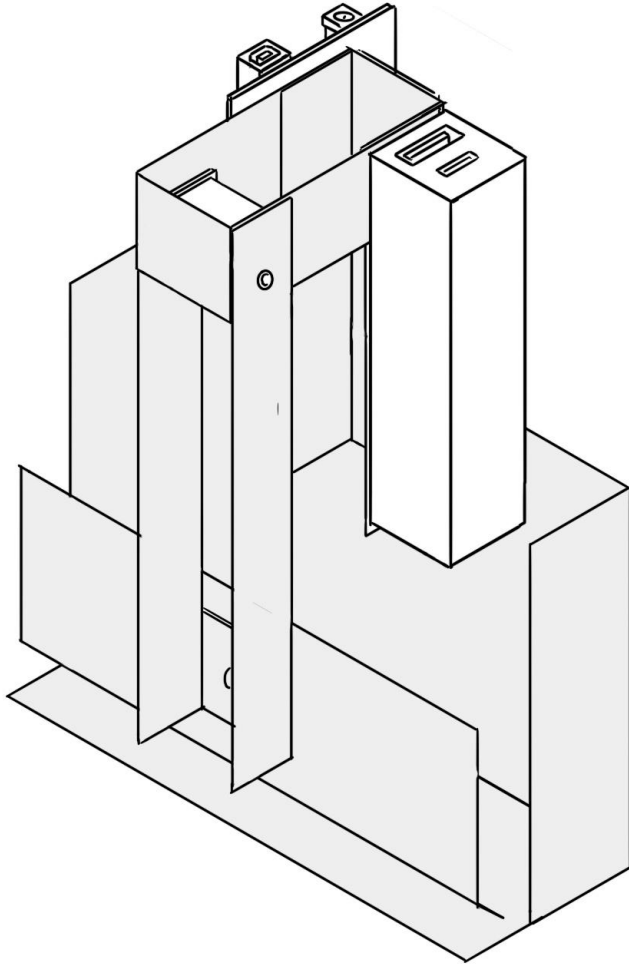
- A new **mounting structure** was integrated into the main body to **support the servo motor** and **provide a stable connection** to the sweeping plate.
- The sweeping plate was attached using a **pivoting linkage** to enable **smoother rotational motion** during operation.



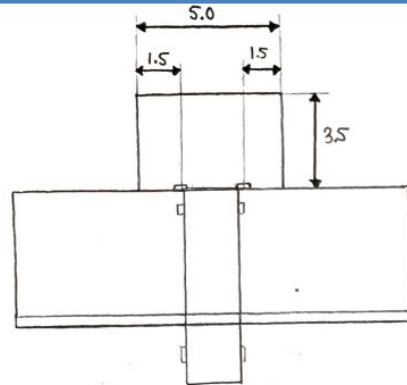
OUR FINAL DESIGN



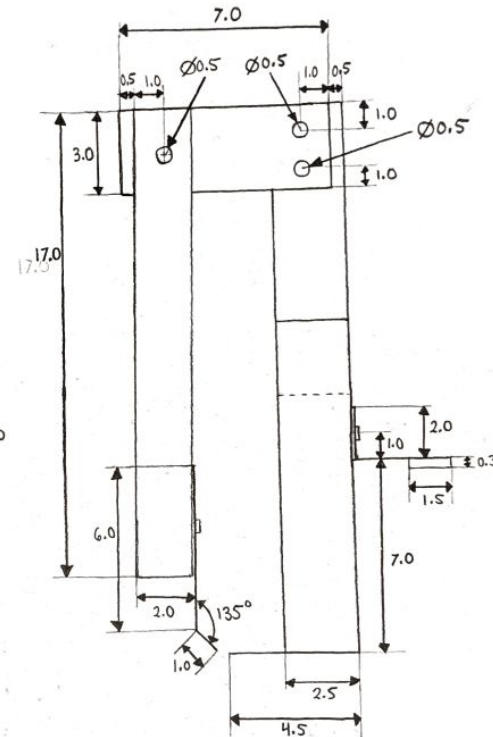
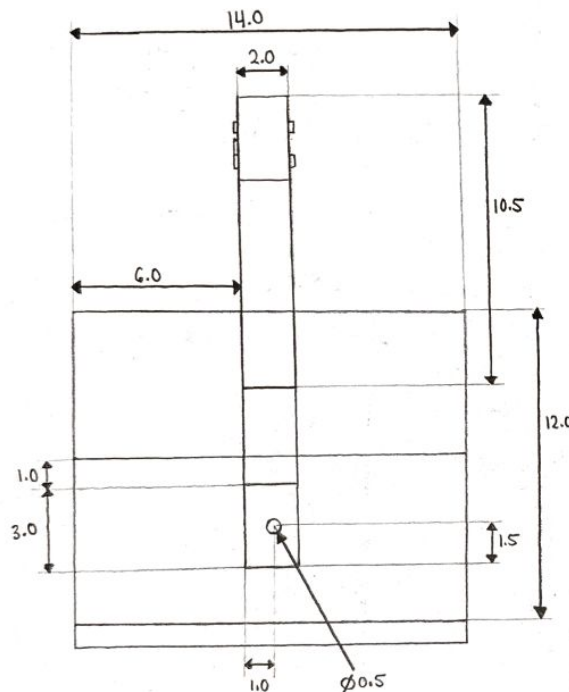
OUR FINAL DESIGN - ISOMETRIC DRAWINGS



OUR FINAL DESIGN - ORTHOGRAPHIC DRAWING



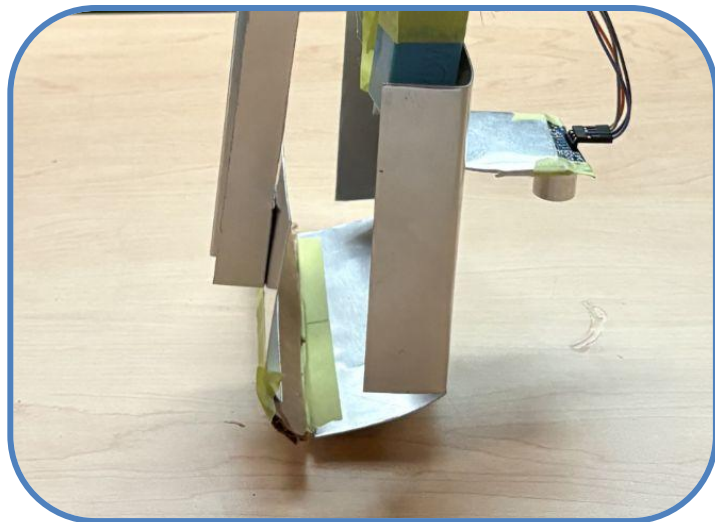
DIMENSIONS:		NAME	DATE	TITLE:
	centimeters (cm)	Drawn M. Edgecll	26/01/20	
Tolerances:		checked E. Goltke	26/01/20	BS Module 5 Claw Orthographic Drawing
	Fractional: ± 0.1 Angular: $\pm 2^\circ$			
		DWG. NO.		REV.
		BS-MS-CLW		A
		SCALE:		1:2



CLAW PERFORMANCE FEATURES

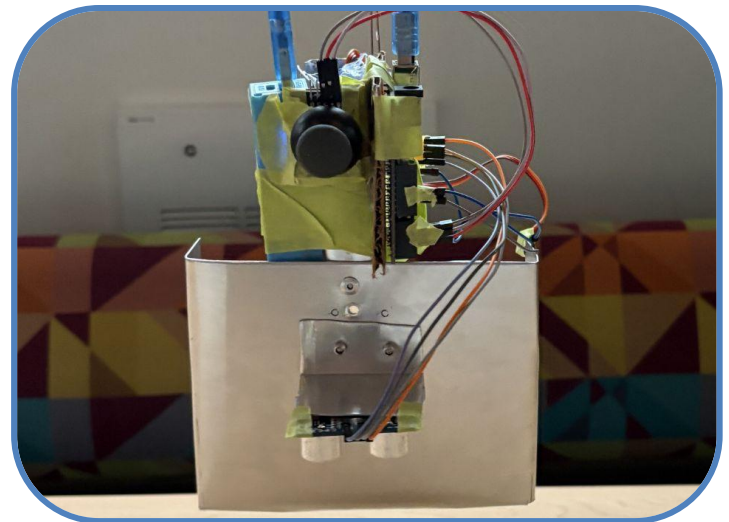
Physical Features

- **Reinforced lip** for carrying longer objects
- **Friction pads** for increased grip
- Spacer from bucket to arm for an **upward sweeping motion**



Arduino & Code

- **Closes** when servo reads 6 cm or less
- **Stays closed** above 20 cm
- **Releases** object when moved below 20 cm
- Emergency **release joystick**



COMPETITION REFLECTION

Reflection

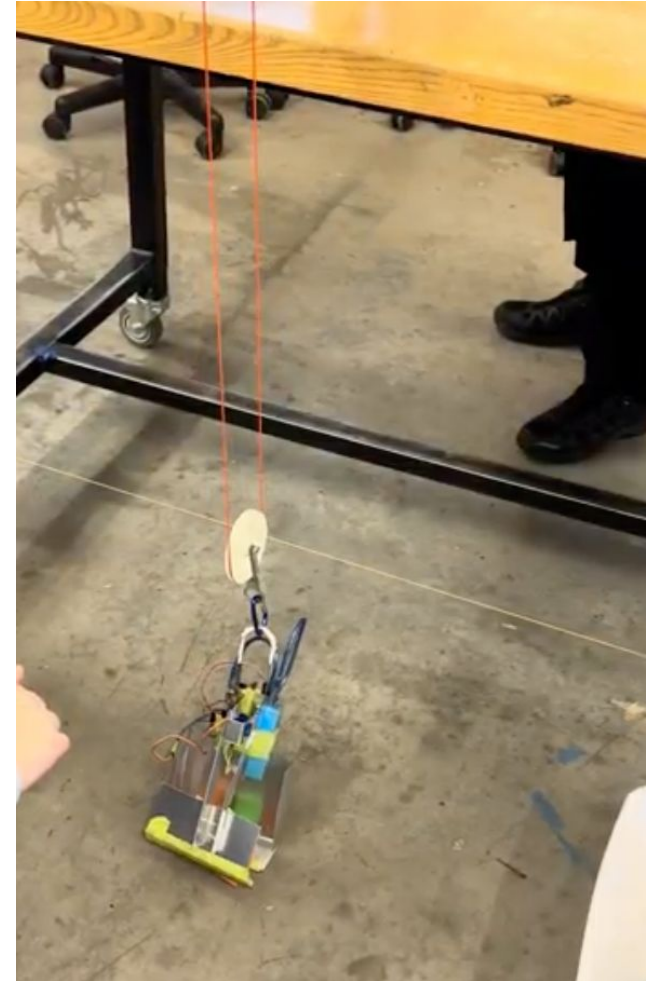
Our team induced maximum potential of the claw with precise description and carefully planned out instructions.

- We created a priority list of objects that are suitable for our claw to pick up
- In competition, we indicated the location of the basket with hand gestures
- Clear and loud instructions, such as: shake, quick drop, keep going down, go above the line, etc.

The objects were also carefully placed for the claw to hold the object(s) at the first try.

Possible Improvements

One thing we could improve is the horizontal swing of the claw when the operator moves the crane sideways. This could be done by moving more slowly or added claw features.



GEN AI ACKNOWLEDGEMENT

Chat GPT 5.2 was used to do a final debugging of our code.

Prompt:

“Think of all scenarios that this code may physically go through, and suggest any debugging changes: [pasted code]”.

Suggestions made concerned potential sonar “bad readings”, fixing timing bugs, using `myservo.write()` more efficiently, and using booleans.



ChatGPT

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